

Rohan Pandey

Bellevue, Washington | (425) 428-2971 | rpande@uw.edu | [Github](#) | [LinkedIn](#) | [Portfolio](#) (Beta) | U.S. Citizen

EDUCATION

University of Washington, Seattle, WA

Expected Graduation: March 2026

BS in Applied and Computational Mathematical Sciences – Scientific Computing and Numerical Analysis

Relevant Coursework: Machine/Reinforcement Learning, Data Structures & Algorithms, Database Systems, High-

Performance Scientific Computing, Optimization (Linear/Nonlinear/Discrete), Data Science, Advanced Neurotechnology

Awards: NASA Space Grant Awardee

SKILLS/TOOLS

Languages: Python, Java, C++, MATLAB, MySQL

Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib

Tools: Git, Linux/Unix, Vim, Bash, OpenBCI, Azure SQL Database, Simulink, Mathematica

Core Competencies: Computer Vision, Deep Learning, Data Structures & Algorithms, Statistical Modeling, Optimization

WORK EXPERIENCE

PricewaterhouseCoopers, Cloud & Digital SAP Intern, Seattle, WA

June – August 2025

- Engineered high-performance Python pipelines for large-scale data ingestion, applying **vectorization, parallel I/O, and memory-efficient batching concepts** from scientific computing to reduce processing latency
- Designed and deployed a secure, **self-service web application** enabling cross-functional teams to configure, trigger, and monitor automated workflows without technical overhead
- Implemented modular, reusable code architecture with robust logging and fault-tolerance, providing a **scalable framework** extensible to ML-driven monitoring and predictive analytics

Fred Hutch, Data Scientist Intern, Seattle, WA

August 2024 – June 2025

- Developed and solved **large systems of nonlinear ODEs** to simulate CAR T-cell and tumor population dynamics, parameterized with patient-derived data.
- Trained regression models with MSE loss to predict therapy outcomes and assess efficacy.
- Integrated nonlinear mixed effects models with numerical solvers to capture immune-cell exhaustion and conditioning dynamics at population and individual levels.

Naval Surface Warfare Center, Reinforcement Learning Intern, Bellevue, WA

October – December 2024

- Developed a reinforcement learning framework using a DDPG agents in MATLAB and Simulink to optimize aerodynamic efficiency in axial turbomachinery
- Simulated complex physical dynamics with embedded control loops and reward functions targeting lift, drag, and energy efficiency tradeoffs
- Integrated streamline curvature methods and potential/Euler flow approximations into a multi-objective design optimization pipeline

MINDCO LABS, Machine Learning Intern, Seattle, WA

September – December 2024

- Built a **real-time computer vision pipeline** for eye-tracking using Python and OpenCV, achieving sub-50ms latency for BCI applications
- Fused multimodal signals by integrating **EEG time-series data with gaze-tracking streams**, feeding into a deep learning pipeline for cognitive-to-text translation
- Experimented with **CNNs, SVMs, and temporal feature extraction methods** for neural signal decoding, establishing baselines for future EEG-to-language models

RESEARCH EXPERIENCE

WASHINGTON eXPERIMENTAL MATHEMATICS LAB

September 2024 – June 2025

- Developed **checksum-triggered backdoors** in MLPs on MNIST, embedding activation logic in pixel parity patterns and masking them with **orthogonal transformations** to remain indistinguishable from clean models under gradient inspection
- Designed **hybrid ReLU activations** and cryptographic-inspired techniques to minimize symmetry, obscure backdoor pathways, and strengthen adversarial ML defenses